

# 1 Solution — GIAM 3.4

## Problem 10

### 1.1 Problem

In an earlier exercise we proved that the quadratic equation  $ax^2 + bx + c = 0$  has two solutions if  $ac < 0$ . Find a counterexample which shows that this implication cannot be replaced with a biconditional.

### 1.2 What Has to Be Refuted

The established theorem is

$$ac < 0 \implies ax^2 + bx + c = 0 \text{ has two solutions.}$$

Turning it into a biconditional would additionally assert the **converse**,

$$ax^2 + bx + c = 0 \text{ has two solutions} \implies ac < 0,$$

so a counterexample is a quadratic with two solutions and  $ac \geq 0$ .

### 1.3 Answer

|                                                                                                |
|------------------------------------------------------------------------------------------------|
| $x^2 - 3x + 2 = 0 : \quad \text{roots 1 and 2,} \quad \text{yet } ac = 1 \cdot 2 = 2 \not< 0.$ |
|------------------------------------------------------------------------------------------------|

### 1.4 Verification

With  $a = 1$ ,  $b = -3$ ,  $c = 2$ :

$$x^2 - 3x + 2 = (x - 1)(x - 2),$$

so the equation has the two distinct solutions  $x = 1$  and  $x = 2$ . Substituting confirms both:  $1 - 3 + 2 = 0$  and  $4 - 6 + 2 = 0$ . Meanwhile

$$ac = 1 \cdot 2 = 2 > 0,$$

so the hypothesis  $ac < 0$  fails while the conclusion holds. The converse is therefore false, and the implication cannot be strengthened to a biconditional. ■

An even more economical witness is  $x^2 + x = x(x + 1)$ , with roots 0 and  $-1$  and  $ac = 0$  — though  $ac = 0$  sits exactly on the boundary, so  $x^2 - 3x + 2$  with  $ac$  strictly positive is the cleaner refutation.

## 1.5 The Picture

Plotting  $b^2$  against  $ac$  shows the whole situation at once: the region where two solutions exist is strictly larger than the region  $ac < 0$ , and the gap between them is where the counterexamples live.

### THE THEOREM (true)

$ac < 0 \Rightarrow ax^2 + bx + c = 0$  has two solutions

### THE CONVERSE (false)

**two solutions  $\Rightarrow ac < 0$**

so the implication cannot be upgraded to a biconditional

### THE COUNTEREXAMPLE

$$x^2 - 3x + 2 = (x - 1)(x - 2)$$

Two solutions,  $x = 1$  and  $x = 2$  — yet  $ac = 1 \cdot 2 = 2$ , which is not negative.

### THE WHOLE SITUATION IN ONE PLOT

A quadratic has two real solutions exactly when  $b^2 - 4ac > 0$ , i.e. above the line  $b^2 = 4ac$ .

#### dark green:

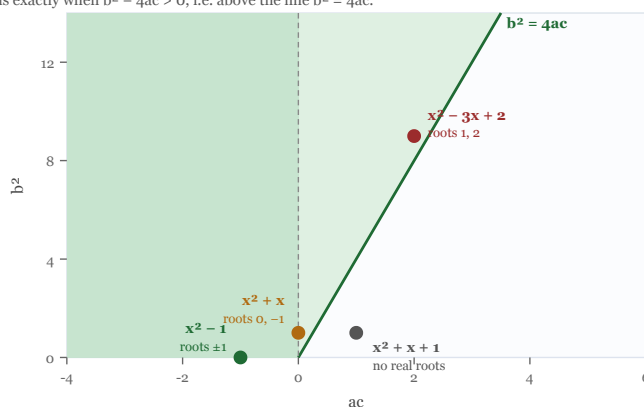
$ac < 0$ .  
All of it lies above  
the line, because  
 $b^2 \geq 0 > 4ac$ .

#### pale green:

also two solutions,  
but  $ac \geq 0$ . Every  
point here refutes  
the converse.

#### white below line:

no real solutions.



### WHAT $ac < 0$ ACTUALLY MEANS

The product of the two roots is  $c/a$ , so  $ac < 0$  says exactly that *the roots have opposite signs* — verified over every  $a, b, c$  from  $-8$  to  $8$ . That is a real restriction: 1 and 2 are both positive.

Geometrically,  $ac < 0$  means the parabola's value at  $x = 0$  has the opposite sign to its arms, so it must cross the axis once on each side of the origin. A parabola crossing twice on the same side is invisible to the test.

### HOW MUCH THE TEST MISSES

Among all integer triples with  $a \neq 0$  and  $-8 \leq a, b, c \leq 8$ , exactly 2832 give two distinct real roots. Of those, 2176 satisfy  $ac < 0$  and 656 do not — so the sufficient condition catches about 77% of them.

**The correct biconditional replaces  $ac < 0$  by the discriminant: two solutions  $\Leftrightarrow b^2 - 4ac > 0$ .**

$ac < 0$  survives as a quick sign test needing no arithmetic;  $b^2 - 4ac > 0$  is exact but must be computed.

Figure 1.1: A diagram in five parts. The first states the true theorem —  $ac < 0$  implies the equation has two solutions — and the false converse — two solutions implies  $ac < 0$  — so the implication cannot be upgraded to a biconditional. The second gives the counterexample:  $x^2 - 3x + 2$  equals the product of  $x - 1$  and  $x - 2$ , which has the two solutions  $x = 1$  and  $x = 2$ , yet  $ac = 1 \times 2 = 2$ , which is not negative. The centre is a plot with  $ac$  on the horizontal axis, running from  $-4$  to  $6$ , and  $b^2$  on the vertical axis, running from  $0$  to  $14$ . A green line rises from the origin representing  $b^2 = 4ac$ . Everything above that line is shaded, marking the region where the quadratic has two real solutions. The left half, where  $ac$  is negative, is shaded darker: all of it lies above the line, because  $b^2$  is at least  $0$  which exceeds  $4ac$ . The pale green wedge between the vertical axis and the line is the region where there are still two solutions but  $ac$  is at least  $0$ , and every point there refutes the converse. The white region below the line is where there are no real solutions. Four points are marked:  $x^2 - 3x + 2$  at  $ac = 2$  and  $b^2 = 9$ , with roots  $1$  and  $2$ , sitting in the pale wedge;  $x^2 + x$  at  $ac = 0$  and  $b^2 = 1$ , with roots  $0$  and  $-1$ ;  $x^2 - 1$  at  $ac = -1$  and  $b^2 = 0$ , with roots  $1$  and  $-1$ , in the dark region; and  $x^2 + x + 1$  at  $ac = 1$  and  $b^2 = 1$ , with no real roots, below the line. A band below explains what  $ac < 0$  actually means: the product of the two roots is  $c/a$ , so  $ac < 0$  says exactly that the roots have opposite signs, verified over every  $a, b$  and  $c$  from  $-8$  to  $8$ . That is a real restriction, since  $1$  and  $2$  are both positive. Geometrically,  $ac < 0$  means the parabola's value at  $x = 0$  has the opposite sign to its arms, so it must cross the axis once on each side of the origin; a parabola crossing twice on the same side is invisible to the test. A final band reports how much the test misses: among all integer triples with  $a$  nonzero and  $a, b, c$  between  $-8$  and  $8$ , exactly  $2832$  give two distinct real roots, of which  $2176$  satisfy  $ac < 0$  and  $656$  do not, so the sufficient condition catches about  $77$  percent of them. The correct biconditional replaces  $ac < 0$  by the discriminant: two solutions if and only if  $b^2 - 4ac > 0$ . The sign test  $ac < 0$  survives as a quick check needing no arithmetic, while the discriminant is exact but must be computed.

The counterexample is the marked point at  $ac = 2$ ,  $b^2 = 9$ : comfortably above the line, so two solutions, but on the wrong side of the vertical axis for the sign test to notice.

## 1.6 Remark — Why It Only Goes One Way

A quadratic with  $a \neq 0$  has two distinct real solutions exactly when its discriminant is positive:

$$\Delta = b^2 - 4ac > 0.$$

If  $ac < 0$  then  $-4ac > 0$ , and since  $b^2 \geq 0$  always,

$$\Delta = b^2 - 4ac \geq 0 + (\text{something positive}) > 0.$$

So  $ac < 0$  is genuinely sufficient — confirmed on every integer triple with  $|a|, |b|, |c| \leq 8$  and  $a \neq 0$ .

But the implication runs only one way, because  $b^2$  can supply the positivity on its own. Whenever

$$0 \leq 4ac < b^2,$$

the discriminant is positive while  $ac$  is not negative. The counterexample sits squarely in that window:  $4ac = 8$  and  $b^2 = 9$ .

## 1.7 Remark — What $ac < 0$ Really Says

The condition has a clean meaning that explains both its usefulness and its limits. For a quadratic with roots  $r_1, r_2$ , Vieta's formulas give

$$r_1 r_2 = \frac{c}{a},$$

so  $ac < 0$  — equivalently  $c/a < 0$  — says precisely that **the two roots have opposite signs**. This was verified over every integer triple in  $[-8, 8]^3$  with  $a \neq 0$ .

That is a real restriction, and it is exactly what the counterexample violates: **1** and **2** are both positive, so their product is positive, so  $ac > 0$  — even though there are certainly two of them.

Geometrically,  $c = f(0)$  and  $a$  controls which way the parabola opens. When  $ac < 0$  the value at the origin has the opposite sign to the arms, so the curve is forced across the axis once on each side of **0**. A parabola that crosses twice on the *same* side of the origin — as  $(x - 1)(x - 2)$  does — has two roots that the sign test simply cannot see.

## 1.8 Remark — How Much the Test Misses

The condition is not a near-miss. Among all integer triples with  $a \neq 0$  and  $-8 \leq a, b, c \leq 8$ :

|                                      |            |
|--------------------------------------|------------|
| triples with two distinct real roots | 2832       |
| of which $ac < 0$                    | 2176       |
| of which $ac \geq 0$                 | <b>656</b> |

So roughly one case in four is a counterexample to the converse. Finding one by random search would have taken a handful of attempts; the difficulty of this exercise is not locating a witness but noticing that the converse is a separate claim at all.

## 1.9 Remark — The Correct Biconditional

A false converse is most useful when it points at the right statement. Replacing  $ac < 0$  with the discriminant gives a genuine biconditional:

$$ax^2 + bx + c = 0 \text{ has two distinct real solutions} \iff b^2 - 4ac > 0 \quad (a \neq 0).$$

The three conditions line up as a strict hierarchy:

| condition       | gives                        | cost to check         |
|-----------------|------------------------------|-----------------------|
| $ac < 0$        | two solutions (one way only) | compare two signs     |
| $b^2 - 4ac > 0$ | two solutions (both ways)    | three multiplications |
| $b^2 - 4ac = 0$ | one repeated solution        | —                     |
| $b^2 - 4ac < 0$ | no real solutions            | —                     |

This is why the weaker statement was worth proving in the first place.  $ac < 0$  can be read off by eye — do  $a$  and  $c$  have opposite signs? — with no arithmetic at all, and when it holds you know immediately that there are two roots straddling the origin. The discriminant is exact but demands computation. A sufficient condition that is cheap to check is often more useful in practice than a necessary and sufficient one that is not; the error lies only in mistaking the first for the second.

## 1.10 Remark — The Shape of the Mistake

This exercise is a specific instance of the most common error in reading mathematical statements. Given a proved implication  $P \Rightarrow Q$ , there are four related conditionals, and only two of them are guaranteed:

| name           | statement                                | status here             |
|----------------|------------------------------------------|-------------------------|
| original       | $ac < 0 \Rightarrow$ two solutions       | true (proved earlier)   |
| contrapositive | no two solutions $\Rightarrow ac \geq 0$ | true — equivalent to it |
| converse       | two solutions $\Rightarrow ac < 0$       | <b>false</b>            |
| inverse        | $ac \geq 0 \Rightarrow$ no two solutions | <b>false</b>            |

The converse and inverse are equivalent to each other and independent of the original, so refuting one refutes both — the single example  $x^2 - 3x + 2$  kills the bottom two rows at once. “Cannot be replaced with a biconditional” is exactly the statement that the third row fails, and  $(x - 1)(x - 2)$  is all the argument required.